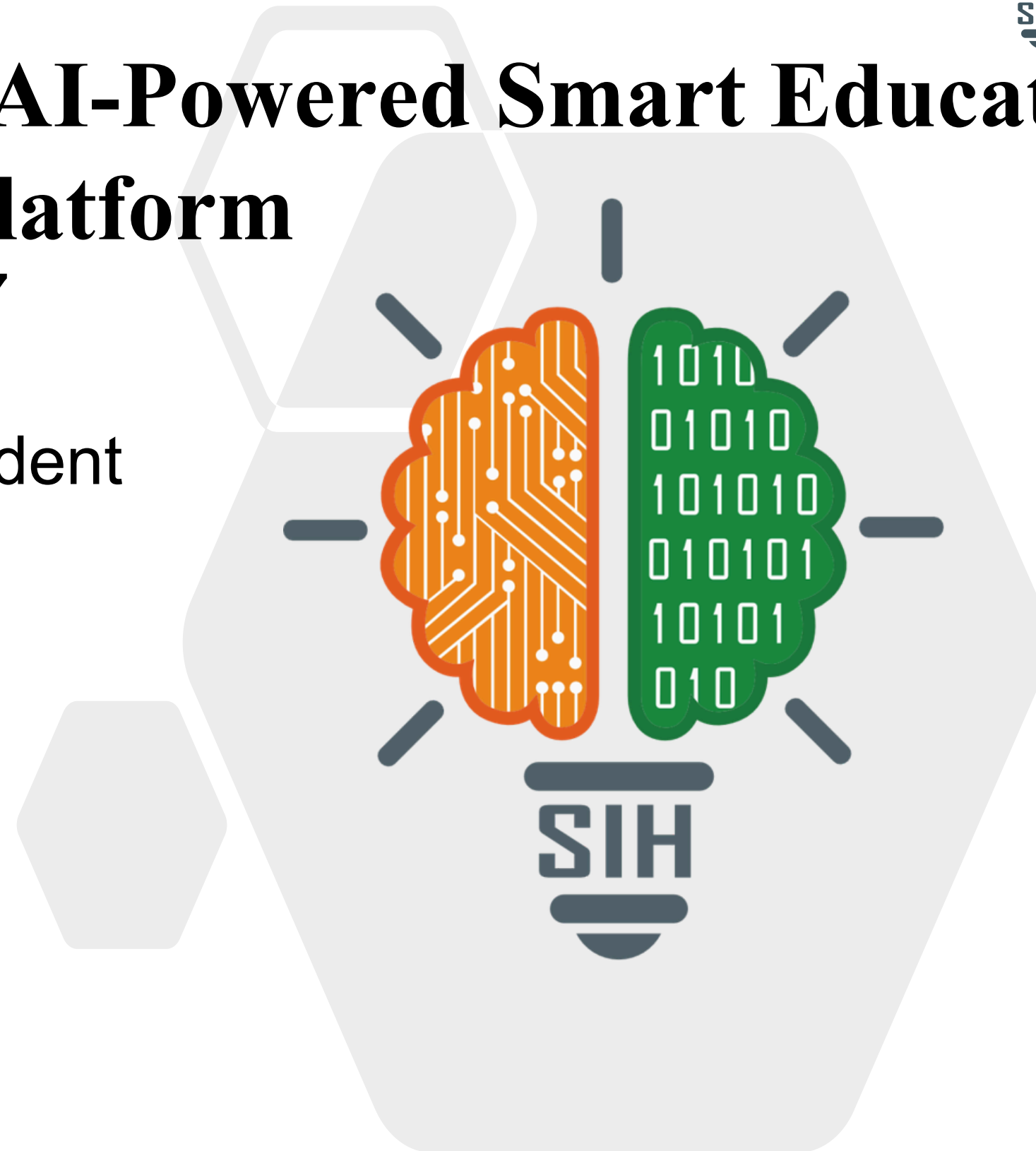


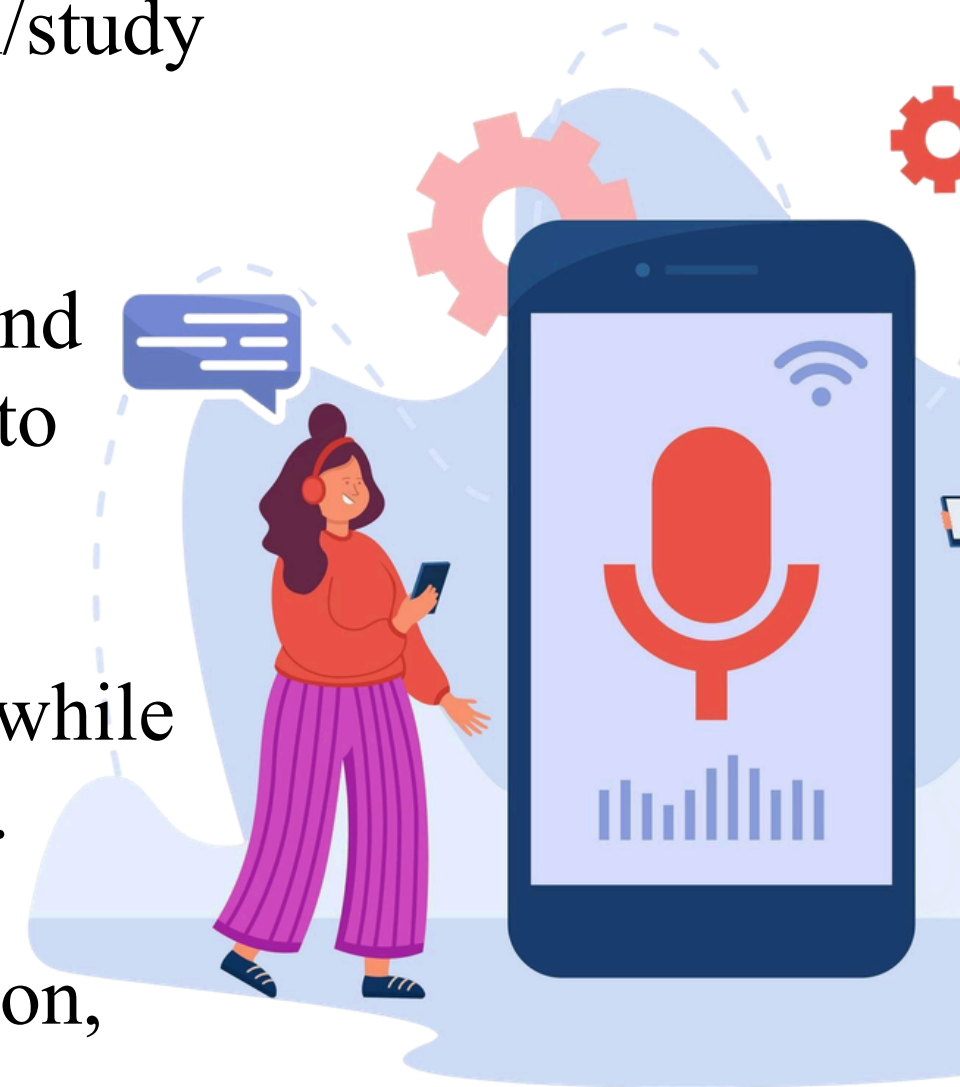
## Talent Management & AI-Powered Smart Education Platform

- Problem Statement ID – SIH26207
- Problem Statement Title- Student Innovation
- Theme- Smart Education
- PS Category- Software
- Team ID-
- Team Name: IQ HOLDERS

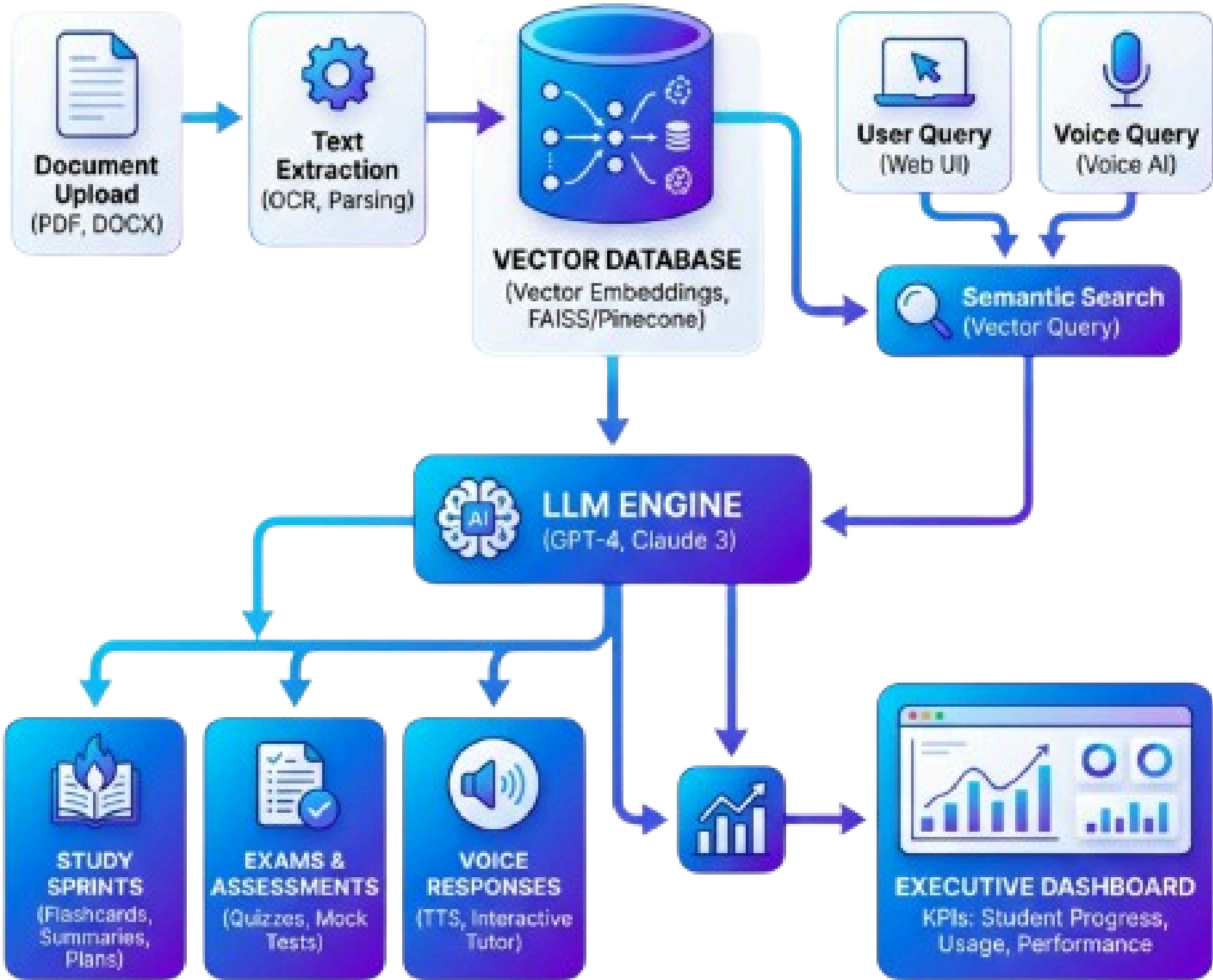


# Talent Management & AI-Powered Smart Education Platform

- **Proposed Solution:** An offline, privacy-first AI platform featuring automated exam/study plan generation, real-time analytics, and an interactive Voice AI Assistant.
- **Detailed explanation:** Uses local LLMs (Ollama) and ChromaDB to ingest PDFs and instantly create 6-day adaptive study sprints and tests, with an executive dashboard to track progress.
- **How it addresses the problem:** Automates manual curriculum design and grading while boosting student engagement through multi-modal learning (visual, text, and voice).
- **Innovation & uniqueness:** 100% offline RAG engine, biometric facial authentication, and a Voice AI that enables hands-free system navigation and 24/7 tutoring.



# TECHNICAL APPROACH

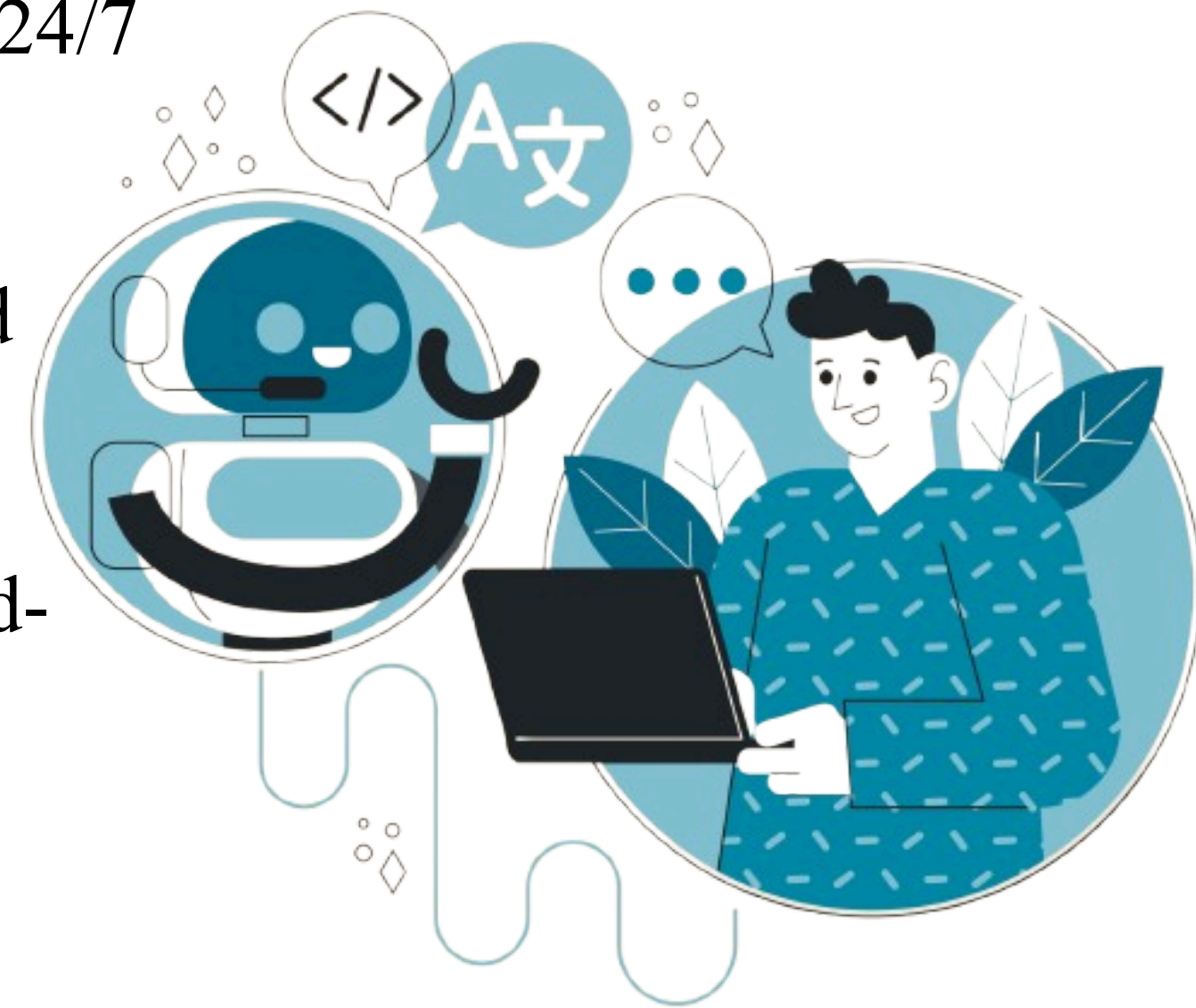


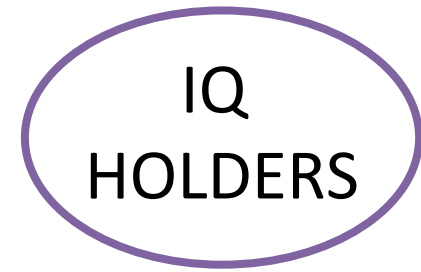
- **Feasibility:** Highly viable and cost-effective. Built on open-source tech (Flask, ChromaDB) with zero recurring API costs due to local AI processing.
- **Potential challenges:** High hardware requirements for local LLMs, and parsing complex PDF layouts.
- **Strategies to overcome:** Use a hybrid approach (Groq API for cloud fallback), optimize document parsers (PyMuPDF), and use background task queues (Celery) to prevent UI lag.



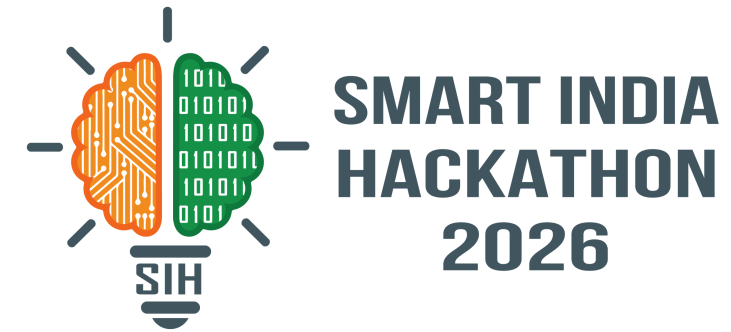


- **Impact on Audience:** Saves educators hundreds of hours in grading/planning while giving students personalized, adaptive 24/7 AI tutoring.
- **Social Benefits:** Democratizes access to tailored education and voice-assisted learning for diverse needs.
- **Economic Benefits:** Eliminates costly LMS licensing and third-party curriculum fees.
- **Security Benefits:** Fully offline architecture ensures sensitive student data never leaves the institution.





# RESEARCH AND REFERENCES



- Research on Retrieval-Augmented Generation (RAG) for hallucination-free education.  
<https://aws.amazon.com/what-is/retrieval-augmented-generation/>
- Use of open-weight models (Ollama/Qwen) for strict Data Privacy.  
<https://ollama.com/>
- Pedagogical studies on Micro-learning & Spaced Repetition (used in our 6-day sprints).  
[Micro-Learning](#)
- Open-source docs: ChromaDB, HuggingFace Sentence-Transformers, and Flask MVC.  
<https://huggingface.co/>